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MULTI-ROBOT SYSTEMS



# Learning an Autonomous Drone Racing Policy via Differentiable Physics

Master's Thesis

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## **Abstract**

**Todo**

**Keywords** Unmanned Aerial Vehicles, Autonomous Drone Racing, Simulation to Reality Transfer, Reinforcement Learning, Domain Randomization

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## Abbreviations

**DR** Domain Randomization

**Sim2Real** Simulation-to-Reality

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## ■ 1 Introduction

### ■ 1.1 Background

**\*Give context about the problem\***

What is ADR?

- Navigate through a sequence of gates as fast as possible within the platform capabilities...

What are the challenges of ADR?

- **Where am I?** → Perception  
challenges: limited sensory data, noisy and uncertain partial observability
- **Where do I go?** → Planning  
safe efficient path planning
- **How do I get there?** → Control  
Handle the non-linear underactuated dynamics to execute the planned motion under actuator limits and model uncertainties

### ■ 1.2 Related Work

Present the state of the art, how do the different research labs approach this problem

#### Open Questions

- How to improve sample efficiency
- Generalization to multiple tasks (how to race while avoiding obstacles, how to race against other drones)
- Transferability of the controllers under unmodeled effects

### ■ 1.3 Objective and Structure

## ■ 2 Preliminaries

### ■ 2.1 Optimal Control Problem

Formulate the optimal control problem, and

### ■ Pontryagin's Maximum Principle

maybe solve it using a pmm and Pontryagin's Maximum Principle

Since it is impossible to perfectly model the state transition equation, and the state estimation is not perfect, the optimal solution breaks under real world conditions.

### ■ 2.2 Classical Control approaches

#### ■ Differential Flatness Based Control

#### ■ Model Predictive Control

### ■ 2.3 Learning-Based approaches

#### ■ Model-Free Reinforcement Learning

#### ■ Model-Based Reinforcement Learning

## ■ 3 Methodology

- 3.1 System Overview
- 3.2 Simulation Framework
  - Quadrotor Model
  - Domain Randomization
  - Supported Control Modes
- 3.3 Training Tasks
- 3.4 Differentiable Optimization

## ■ 4 Results

- 4.1 Evaluation Criteria
- 4.2 Baseline Performance
- 4.3 Ablation Study
- 4.4 Transferability to MRS UAV System
- 4.5 Real World Validation

## ■ 5 Conclusion

This work investigated the effectiveness of Domain Randomization (DR) in bridging the Simulation-to-Reality (Sim2Real) gap for learning-based quadcopter racing controllers. By varying the randomization level of key drone physical parameters, we demonstrated that DR significantly improves policy robustness and transferability across different vehicle platforms.

Our simulated experiments revealed trade-off between performance and reliability. The baseline policy trained without randomization (0% DR) achieved the fastest lap times of 4.13 s but suffered from poor stability, with over 50% of laps resulting in gate collisions and two fatal crashes among the three tested drone instances. In contrast, policies trained with moderate randomization (10% and 20% DR) exhibited substantially improved success rates exceeding 97%, while maintaining competitive lap times of 5.73 s and 5.38 s respectively. The 20% DR configuration emerged as the optimal balance, achieving a perfect 100% success rate across all three simulated drones without sacrificing significant performance.

However, our results also indicate diminishing returns and potential instability at higher randomization levels. The 30% DR policy showed degraded performance, particularly for UAV3, suggesting that excessive randomization can destabilize the learning process and reduce sample efficiency. This aligns with our hypothesis that the randomization space must be carefully bounded to encompass the real domain characteristics without introducing unnecessary variability that hinders convergence.

## ■ 6 Future Directions

## ■ 7 References

## ■ A Appendix A

Next table presents the parameter multipliers ( $\ast \mathcal{U}(1 - p, 1 + p)$ ) used to randomize the three simulated drones in Gazebo.

| Parameter                   | Nominal     | Randomized                               |               |               |
|-----------------------------|-------------|------------------------------------------|---------------|---------------|
|                             |             | Nominal $\ast \mathcal{U}(1 - p, 1 + p)$ |               |               |
|                             |             | uav1<br>p = 10%                          | uav2<br>p=20% | uav3<br>p=30% |
| mass [kg]                   | 1.178       | 1.041                                    | 0.814         | 1.212         |
| body_radius [m]             | 0.05        | 0.964                                    | 0.883         | 0.702         |
| body_height [m]             | -0.1        | 1.023                                    | 1.177         | 1.271         |
| mass_prop [kg]              | 0.008       | 1.095                                    | 1.143         | 1.191         |
| radius_prop [m]             | 0.089408    | 1.072                                    | 0.918         | 1.288         |
| motor_mesh_z_offset [m]     | -0.0086     | 0.924                                    | 0.824         | 0.924         |
| rotor_xy_offset [m]         | 0.1         | 1.067                                    | 1.192         | 1.067         |
| rotor_z_offset [m]          | -0.055      | 0.912                                    | 1.118         | 0.912         |
| rotor_velocity_slowdown_sim | 0.0159236   | 1.034                                    | 1.132         | 0.713         |
| motor_constant              | 20.21       | 1.081                                    | 0.804         | 1.229         |
| moment_constant             | 0.012       | 0.922                                    | 1.162         | 0.731         |
| time_constant_up [s]        | 1/(20.5761) | 0.953                                    | 0.886         | 1.221         |
| time_constant_down [s]      | 1/(18.3486) | 1.067                                    | 1.189         | 0.708         |
| max_rot_velocity            | 1.0         | 1.058                                    | 0.833         | 1.197         |
| rotor_drag_coefficient      | 0.1         | 1.041                                    | 1.144         | 1.276         |
| inertia_body_radius [m]     | 0.25        | 1.091                                    | 0.817         | 1.298         |
| inertia_body_height [m]     | 0.05        | 0.912                                    | 1.178         | 0.744         |

Table A.1: Parameter multipliers for 10%, 20%, and 30% randomized UAVs. Nominal column retains original values.